

CelebA: IID

COMPLETED VALIDATION COHORT - 10 shared seeds (91001-91010)

Accepted snapshot: 700/700 records, 2026-09-28T10:20:44 UTC

Data distribution			IID ($\alpha = 5000$)				
Category	Method	Metric	Benign	F-Flip	FedSA	S-DFA	Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	84.72 $\pm$ 11.74	89.57 $\pm$ 0.78	68.19 $\pm$ 5.77	66.01 $\pm$ 3.57	84.37 $\pm$ 2.38
		AEOD $\downarrow$	0.0384 $\pm$ 0.0176	0.0519 $\pm$ 0.0096	0.1004 $\pm$ 0.0410	0.0950 $\pm$ 0.0477	0.0280 $\pm$ 0.0076
		ASPD $\downarrow$	0.0937 $\pm$ 0.0366	0.1212 $\pm$ 0.0064	0.0993 $\pm$ 0.0380	0.0807 $\pm$ 0.0413	0.0693 $\pm$ 0.0161
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.47 $\pm$ 0.85	89.79 $\pm$ 0.76	69.47 $\pm$ 4.65	62.59 $\pm$ 5.11	85.55 $\pm$ 2.31
		AEOD $\downarrow$	0.0414 $\pm$ 0.0076	0.0489 $\pm$ 0.0042	0.1158 $\pm$ 0.0635	0.1594 $\pm$ 0.0822	0.0288 $\pm$ 0.0081
		ASPD $\downarrow$	0.1102 $\pm$ 0.0077	0.1188 $\pm$ 0.0043	0.1120 $\pm$ 0.0554	0.1461 $\pm$ 0.0668	0.0781 $\pm$ 0.0116
Robust FL	Median	ACC (%) $\uparrow$	86.28 $\pm$ 1.31	86.52 $\pm$ 1.35	84.66 $\pm$ 1.47	85.00 $\pm$ 1.38	85.54 $\pm$ 1.50
		AEOD $\downarrow$	0.0510 $\pm$ 0.0081	0.0610 $\pm$ 0.0087	0.0438 $\pm$ 0.0135	0.0484 $\pm$ 0.0090	0.0534 $\pm$ 0.0096
		ASPD $\downarrow$	0.0980 $\pm$ 0.0117	0.1062 $\pm$ 0.0111	0.0825 $\pm$ 0.0142	0.0852 $\pm$ 0.0127	0.0965 $\pm$ 0.0124
Robust FL	FLTrust	ACC (%) $\uparrow$	89.43 $\pm$ 0.75	89.67 $\pm$ 0.82	89.68 $\pm$ 0.74	89.68 $\pm$ 0.74	89.72 $\pm$ 0.64
		AEOD $\downarrow$	0.0432 $\pm$ 0.0075	0.0518 $\pm$ 0.0072	0.0422 $\pm$ 0.0069	0.0422 $\pm$ 0.0069	0.0496 $\pm$ 0.0061
		ASPD $\downarrow$	0.1111 $\pm$ 0.0062	0.1207 $\pm$ 0.0064	0.1119 $\pm$ 0.0071	0.1119 $\pm$ 0.0071	0.1177 $\pm$ 0.0057
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.18 $\pm$ 0.83	89.57 $\pm$ 1.24	49.32 $\pm$ 1.59	50.33 $\pm$ 1.72	83.20 $\pm$ 1.76
		AEOD $\downarrow$	0.0317 $\pm$ 0.0026	0.0369 $\pm$ 0.0071	0.0042 $\pm$ 0.0132	0.0000 $\pm$ 0.0000	0.0205 $\pm$ 0.0098
		ASPD $\downarrow$	0.0998 $\pm$ 0.0062	0.1078 $\pm$ 0.0092	0.0049 $\pm$ 0.0156	0.0000 $\pm$ 0.0000	0.0491 $\pm$ 0.0199
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.93 $\pm$ 1.10	90.37 $\pm$ 0.60	77.83 $\pm$ 12.15	75.99 $\pm$ 16.52	86.44 $\pm$ 5.76
		AEOD $\downarrow$	0.0365 $\pm$ 0.0075	0.0363 $\pm$ 0.0086	0.0298 $\pm$ 0.0299	0.0293 $\pm$ 0.0146	0.0368 $\pm$ 0.0144
		ASPD $\downarrow$	0.1074 $\pm$ 0.0113	0.1118 $\pm$ 0.0100	0.0607 $\pm$ 0.0354	0.0652 $\pm$ 0.0419	0.0903 $\pm$ 0.0325
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.26 $\pm$ 1.04	88.39 $\pm$ 0.63	88.47 $\pm$ 0.91	88.69 $\pm$ 0.81	88.38 $\pm$ 0.89
		AEOD $\downarrow$	0.0097 $\pm$ 0.0075	0.0107 $\pm$ 0.0096	0.0061 $\pm$ 0.0042	0.0077 $\pm$ 0.0046	0.0165 $\pm$ 0.0072
		ASPD $\downarrow$	0.0625 $\pm$ 0.0138	0.0607 $\pm$ 0.0103	0.0640 $\pm$ 0.0142	0.0638 $\pm$ 0.0070	0.0559 $\pm$ 0.0156

All 70 method/distribution/scenario cells use the same 10 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.

Round 70; validation only (19,867 images). Mean $\pm$ sample SD (ddof = 1); ACC in %, gaps on [0, 1].

AEOD denotes the implemented absolute TPR gap. No significance claim follows from mean ranks alone.

* Project adaptations. GuardFed includes training-root group calibration; baseline outputs are uncalibrated.

Recipes were selected on non-IID Benign/S-DFA, including seed 91001; this is coverage transfer, not IID-specific tuning.

All records in this table use torch 2.11/cu128. The source cohort contains 14 cu130 records, excluded from this table.

Stage A covers seven implementations; ten further manuscript baselines and mechanism controls remain outstanding.